# IDS 702 Project Proposal

Haoran Wang, Laura Perkins, Masoma Shoayb

## Project Proposal/Statistical Analysis Plan

Link to your team's Github Repo:

### Dataset Option 1

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Project Title:

Data citation (use APA formatting or something similar):

Original data collection date (when was the dataset published?):

Data pull date (when did you download the data you will use?):

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### 1 Research Question

*Write the research question you are interested in investigating in your analysis. Consult this helpful guide for writing a good research question. Go beyond "is there a relationship between X and Y" to incorporate some statistical complexity, e.g., "is there a relationship between X and Y and does this relationship change over time?" You may also want to provide a second RQ as a backup and I can provide feedback on both options. Remember that you are required to use an inferential research question.*

### 2 Study Population

Describe the observation units and how the data were collected.

## 3 Outcomes, Exposures, and Additional Variables of Interest

*Add rows as necessary. Use the data dictionary for the coding specifications, e.g., 1=Diagnosed with CHD, 0=No diagnosis*

3.1 Primary Outcome(s)

| Variable Name | Description | Coding specification |
|---|---|---|
| | | |

3.2 Primary Independent Variable(s)

| Variable Name | Description | Coding specification |
|---|---|---|
| | | |

3.3 Additional Variables of Interest (including potential confounders)

| Variable Name | Description | Coding specification |
|---|---|---|
| | | |

## 4 Statistical Analysis Plan

4.1 Exploratory Data Analysis

*Detail how you will investigate the data, including basic characteristics of the outcome and independent variable(s) of interest (e.g., "The outcome is continuous, so we will generate a histogram of the outcome. Then, we will generate a stacked histogram/scatter plot to explore the relationship between the outcome and the independent variable")*

4.2 Statistical Model (write in mathematical notation and define each term)

4.2 Rationale for the statistical model (why are you choosing to use this type of model? What about your data structure/characteristics are you accommodating with this model type?)

4.3 Model Assumptions

4.4 Model Assessment (how will you assess the assumptions and model performance?)

## Exploratory Data Analysis

*First, provide a snapshot of the data using the glimpse function or something similar. Then, present your EDA according to your SAP above. At minimum, you should present basic characteristics of your outcome alone (descriptive statistics and a plot) and at least one key relationship of interest, but the EDA should not exceed 3 pages. It may be useful to present a "Table 1" that gives descriptive statistics for several variables of interest. You are not required to complete data cleaning for the EDA.*

**Dataset Option 2**

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Project Title:

Data citation (use APA formatting or something similar):

Original data collection date (when was the dataset published?):

Data pull date (when did you download the data you will use?):

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## 1 Research Question

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| | | |

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## Team Charter

1. When will you meet as a team to work on the project components? Will these meetings be held in person or virtually?
2. What is your group policy on missing team meetings (e.g., how much advance notice should be provided)?
3. How will your team communicate (email, Slack, text messages)? What is your policy on appropriate response time (within a certain number of hours? Nights/weekends?)?